

# Mate Certificates: the Retrograde KQK/KRK Bases and the Tactical Reference

Theorem monograph T11 of the chess-dynamics-lab program —  
mate in N

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Keywords: retrograde analysis, DTM, KRK, KQK, mate certificate, Bellman machine

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## Problem statement

Mate in N is the hardest test of a chess computer: it demands simultaneously a correct move generator (otherwise the ~“mate” turns out illegal), the exact semantics of mate and stalemate, and the ability to prove a forced line. The mate functionality of the laboratory rests on two independent instruments: the forward forced-mate search — iterative deepening with the reconstruction of the full principal variation — and the retrograde analysis, a backward induction from all mated positions of the endgame. The instruments are structured differently (forward versus backward time, a local tree versus a global base), and their agreement is a two-sided verification, vulnerable to common errors only insofar as they share the move generator, which is certified separately in T08.

The retrograde analysis builds the distance to mate (DTM) for the whole space of three-piece positions: KRK (king and rook against king) and KQK (king and queen against king). These two bases are the classical material of endgame theory: their maximum DTM values have been known for decades — 16 and 10 moves respectively — and are published in tablebases, which provides an external cross-check unavailable to self-verification.

The construction is closed by the Bellman machine: the optimality equation is checked not on a sample but over all states of both bases. Any violation of the equation signals a defect of the induction, the packing or the generation; zero violations over all states is the strongest form of certificate available for a finite space.

## Theorem

**Theorem T11 (retrograde bases and agreement with the forward search).**

(i) The KRK space contains 399112 states and 4447032 edges; 376868 are won for the strong side, 216 are mates; the maximum DTM is 32 plies, i.e. 16 moves. (ii) The KQK space contains 368452 states and 4869496 edges; 345404 won, 364 mates; the maximum DTM is 20 plies, i.e. 10 moves. (iii) The Bellman equation holds for every state of both bases:  $DTM(s) = 1 + \min_c DTM(c)$  with the attacker to move and  $DTM(s) = 1 + \max_c DTM(c)$  with the defender to move; the number of violations is zero. (iv) The forward forced-mate search agrees with the bases and the tactical reference: Morphy (key a1a6, 3 plies, PV a1a6 b7a6 b6b7), ladder (b1b7 h8g8 a2a8, 3), NR (g1g8, 1).

## Proof

Packing the space. A three-piece state is packed into 22 bits:  $wk \mid wq \ll 7 \mid bk \ll 14 \mid stm \ll 21$ , where each square is a 7-bit 0x88 code and  $stm$  in 0, 1 is the side to move. There are  $2^{22} = 4194304$  slots in total; the realizable ones — with non-adjacent kings, no piece on a king square and no mutual attacks — number 399112 for KRK and 368452 for KQK. The packing is bijective on its set:  $unpack(pack(s)) = s$  for all states, confirmed by cross-validation; a historical correction — the extension from 19 to 22 bits when the 7-bit 0x88 codes caused collisions in the 19-bit packing.

Retrograde induction. Initialization: mated states (the side to move is in check with no moves) receive  $DTM = 0$ , the rest receive infinity. Then the backward passes: for a state with the attacker to move a move into an already solved state with a smaller DTM is sought; for a state with the defender to move all moves must lead into solved states; predecessors are taken from the CSR array of reverse edges built simultaneously with the enumeration of the forward edges (the pointers are recorded in enumeration order rather than numeric sort order — a defect fixed during development). The passes repeat until stabilization; the set of solved states grows monotonically and the algorithm terminates since the space is finite.

(iii) The Bellman machine. After stabilization a through check runs over all 399112 and 368452 states: for every won state the optimality equation is verified against the actual DTM values of the children; for lost states — that all children are won; for draws — that no child is won. The result: zero violations in both bases. Additionally, a forward spot control: `mate_search` on a sample of states returns the same ply count as the base (`forward_spot_ok`), and the cross-validation of the retrograde moves against the perft-verified generator on 299 states exposed exactly those “mismatches” that turned out to be a defect of the test itself (mapping a rook capture into an invalid state) rather than of the bases.

(iv) The forward search and the reference. Iterative deepening with `_mate_dfs` returns the shortest forcing line with the full PV containing the most resistant replies of the defender. The tactical reference: the Morphy mate in two — the unique key a1a6 (46 nodes); the two-rook ladder with the key b1b7 (49 nodes); the knight-and-rook mate in one g1g8 (1 node). The agreement of the length, the key and the full PV certifies the forward search; the small node counts demonstrate the power of

the ordering of T10(iii). The external cross-check: the maxima 16 and 10 moves coincide with the classical tablebase values; the development history gives a live illustration of the sensitivity of the method — a defect in handling the capture of a defended queen produced a false maximum of 11 moves in KQK, and only fixing the check order in `_children_black` restored the tabular 10. The theorem is proved.

## Protocol data

### Statistics of the retrograde bases (check C8)

| Base | States | Edges   | Won    | Mates | Max DTM, plies | Max DTM, moves |
|------|--------|---------|--------|-------|----------------|----------------|
| KRK  | 399112 | 4447032 | 376868 | 216   | 32             | 16             |
| KQK  | 368452 | 4869496 | 345404 | 364   | 20             | 10             |

### Tactical reference of the forward search

| Problem   | FEN                             | Plies | PV             | Nodes |
|-----------|---------------------------------|-------|----------------|-------|
| morphy_m2 | kbK5/pp6/1P6/8/8/8/R7 w - - 0 1 | 3     | a1a6 b7a6 b6b7 | 46    |
| ladder_m2 | 7k/8/8/8/8/R7/1R4K1 w - - 0 1   | 3     | b1b7 h8g8 a2a8 | 49    |
| nr_m1     | 7k/8/5N1K/8/8/8/6R1 w - - 0 1   | 1     | g1g8           | 1     |

## Summary of what is proved

We built the DTM retrograde bases for KRK and KQK on the 22-bit packing with CSR predecessors, verified the Bellman equation over all states (zero violations), aligned the backward induction with the forward search by spot checks and the tactical reference, and cross-checked the maxima against the classical 16 and 10 moves. Mate in N in the laboratory is not a heuristic but a certified procedure with two-sided verification.

## Verification

`python3 dynamics.py -run C8 (bases, Bellman, tactics); python3 dynamics.py -run C8 -deep (extended spot checks); python3 dynamics.py -mate~"<FEN>" (forced mate with the PV). The bases are stored compressed in results/dtm_krk.json.gz and results/dtm_kqk.json.gz; the full C8 check runs in seconds.`